9.4.46

EE25BTECH11033 - Kavin

Question:
A pole has to be erected at a point on the boundary of a circular park of diameter 1.3 metres in such a way that the difference of its distances from two diametrically opposite fixed gates A and B on the boundary is 7 metres. Is it possible to do so? If yes, at what distances from the two gates should the pole be erected?

Solution:

Let the circular park have gates **A** and **B** at the ends of a diameter. The distance between them, AB, is the diameter, which is 13 m. Let the pole **P** be a point on the boundary of the circle.

- The angle subtended by the diameter at any point on the circumference is a right angle (90°). Therefore, $\triangle APB$ is a right-angled triangle with the hypotenuse being the diameter AB.

Using Baudhayana's theorem for $\triangle APB$

$$PA^2 + PB^2 = AB^2$$

$$PA^2 + PB^2 = 169 \tag{1}$$

We are also given that the difference of the distances from the pole to the gates is 7 meters:

$$|PA - PB| = 7 \tag{2}$$

Assuming $PA > PB$, we have:

$$PA - PB = 7 \tag{3}$$

$$\implies PB = PA - 7 \tag{4}$$

$$\implies PA^2 + (PA - 7)^2 = 169 \tag{5}$$

Let PA be x,

$$\implies x^2 + (x - 7)^2 = 169 \tag{6}$$

$$\implies y = x^2 - 7x - 60 = 0 \tag{7}$$

which can be expressed as the conic

$$\mathbf{x}^\top \mathbf{V}\mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \tag{8}$$

$$\mathbf{V} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \mathbf{u} = \begin{pmatrix} -\frac{7}{2} \\ -\frac{1}{2} \end{pmatrix}, f = -60. \tag{9}$$

To find the roots of the expression, we find the points of intersection of the conic with the x-axis

$$\mathbf{x} = \mathbf{h} + k\mathbf{m} \tag{10}$$

$$\mathbf{h} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{m} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{11}$$

The points of intersection of the line

$$L : \quad \mathbf{x} = \mathbf{h} + \kappa\mathbf{m} \quad \kappa \in \mathbb{R} \tag{12}$$

with the conic section is given by

$$\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m} \tag{13}$$

where

$$\kappa_i \quad = \quad \frac{1}{\mathbf{m}^\top \mathbf{V}\mathbf{m}} \left(-\mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u}) \quad \pm \quad \sqrt{[\mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u})]^2 - \mathrm{g}(\mathbf{h}) (\mathbf{m}^\top \mathbf{V}\mathbf{m})} \right) \tag{14}$$

The values of k are given by

$$k_i = \frac{1}{1} \left(\frac{7}{2} \pm \sqrt{\left(\frac{7}{2}\right)^2 + 60} \right) \tag{15}$$

$$\implies k_1 = -5, k_1 = 12 \tag{16}$$

Hence the points of intersection are

$$\mathbf{h} + k\mathbf{m} = \begin{pmatrix} -5 \\ 0 \end{pmatrix}, \begin{pmatrix} 12 \\ 0 \end{pmatrix} \tag{17}$$

Hence the solutions are $x = -5$ and $x = 12$. We reject $x = -5$ as the length of the side cannot be negative.
This gives the value $PA = 12$ and $PB = 5$.

Therefore, it is possible to erect the pole as described.

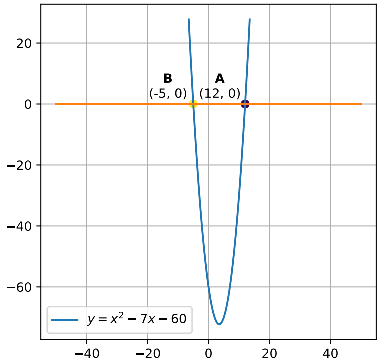

Fig. 0: Intersection of $y = x^2 - 7x - 60$ with the x-axis